BAMS-FE: Band-by-Band Adaptive Multiscale Superpixel Feature Extraction for Hyperspectral Image Classification

Jianmeng Li, Hui Sheng, Mingming Xu, *Member, IEEE*, Shanwei Liu, and Zhe Zeng

***Abstract*— Superpixel segmentation has emerged as a prominent approach for simultaneous extraction of spatial–spectral features in hyperspectral imagery, exhibiting considerable efficacy in this domain. Although effective in spatial spectrum feature extraction, the existing feature extraction algorithms typically perform superpixel segmentation on a single band, failing to utilize the rich spectral and spatial information available across more bands. Moreover, current superpixel feature extraction methods lack scientific guidance for determining optimal multiscale parameters, which can lead to suboptimal segmentation and increased the complexity of hyperspectral analysis. To overcome these limitations, this article presents a novel band-by-band adaptive multiscale superpixel feature extraction (BAMS-FE) method. The method comprises of two key components: a band-by-band superpixel-based feature extraction method and an adaptive optimal superpixel multiscale determination method. First, the band-by-band superpixel-based feature extraction method performs superpixel segmentation for each band of hyperspectral images (HSIs), thereby extracting joint spatial and spectral features. Second, the adaptive optimal superpixel multiscale determination method uses an unsupervised approach to determine the optimal multiscale superpixel segmentation parameters. Finally, the BAMS algorithm is obtained by combining the above two algorithms. The proposed algorithm is evaluated on five different datasets, and the results demonstrate its excellent precision and stability. With the top 99% principal components post-PCA transformation or with raw, unprocessed hyperspectral datasets, stable and satisfactory classification performance is achieved by BAMS. Additionally, we compared its performance with several other state-of-the-art algorithms and found that it outperformed them in terms of accuracy. Our code is publicly available at https://github.com/UPCGIT/BAMS-FE.**

***Index Terms*— Band-by-band, feature extraction, hyperspectral image (HSI), multiscale, superpixel.**

I. Introduction

HYPERSPECTRAL image (HSI) has become increasingly popular in recent years due to its ability to capture a large number of spectral bands with narrow spectral ranges [1], [2], [3], [4]. Compared to multiband images, HSIs offer advantages for complex Earth object classification, making them widely applicable in various fields, such as military, agriculture, and environment [5], [6], [7].

However, as the dimensionality of HSIs increases, the classification performance of a classifier may not necessarily improve, and the opposite effect, known as the Hughes phenomenon, may occur [2]. In addition, the hyperspectral imaging process is susceptible to various interfering factors, resulting in spectral variability [8]. To address these challenges, various spectral feature extraction algorithms, such as principal component analysis (PCA) [9], linear discriminant analysis (LDA) [10], and independent component analysis (ICA) [11], have been combined with classification methodologies, such as support vector machine (SVM) [12] and random forest (RF) [13] for HSI classification tasks, resulting in satisfactory performance.

To further improve the effectiveness of feature extraction, many researchers have applied spatial–spectral joint feature extraction techniques to classification tasks in HSIs. In [14], a set of Gabor wavelets with different frequencies and orientations is proposed to extract distinct features in both the spatial and spectral domains. Another approach involved designing different types of convolutional neural networks that leveraged the powerful feature extraction capability of convolutional kernels in both the spatial and spectral dimensions to enhance the accuracy of HSI classification. These networks are designed in various studies, such as [15], [16], [17], [18], [19], and [20]. Moreover, the classification task based on spatial–spectral joint feature extraction can also be accomplished by adopting networks with encoders [21], transformers [22], or graph convolutional network [23].

Despite the efficacy of the aforementioned methods, they suffer from a common drawback, as the spatial feature extraction range is often fixed in square windows, thereby disregarding the diverse spatial structure information that exists in remote sensing imagery [24]. To address this limitation, superpixel segmentation has been introduced to cluster similar pixels in adjacent regions into homogenous regions with adaptive shapes that align with the assumption that objects in close proximity to one another in geographical space are more likely to belong to the same object class. Superpixels have thus overcome the deficiency of fixed-shape spatial feature extraction methods and have led to

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a body of research that has yielded promising results in HSI classification. For example, in [25], a framework named flexible Gabor-based superpixel-level unsupervised LDA (FG-SuULDA) is proposed to extract the most informative and discriminating features for classification. Similarly, in [26], a novel local binary pattern (LBP)-based superpixel-level decision fusion method is proposed for HSI classification. In [27] and [28], discriminant analysis and kernel methods are, respectively, used in combination with superpixels to extract spatial–spectral features from HSIs. In [29] and [30], the classification of HSIs is studied using sparse representation based on superpixels. Given that superpixel segmentation can reduce the spatial resolution of remote sensing images from pixel level to region level, preprocessing method based on superpixels has found wide application in graph-based classification models [31], [32], [33], [34].

The superpixel-based feature extraction technique mentioned above typically employs a fixed scaling. However, the utilization of superpixels of different scales can efficiently extract information from various spatial constructs and is often better than a single superpixel scaling algorithm. For instance, multiscale segmentation-based superpixelwise PCA (MSuperPCA) [35], which aggregates results from multiple superpixelwise PCA (SuperPCA) votes, outperforms a single-size SuperPCA. In [24], not only the features of each scale are captured through the superpixel-based approach, but the correlation between different scales is also considered via a recurrent neural network. In [36] and [37], the multiscale superpixel method and representation-based method are applied to classification tasks, respectively. Furthermore, in [3], [16], [38], [39], [40], and [41], subpixel, pixel, and superpixel multiscale feature fusion are utilized in HSI classification.

Despite the remarkable progress made in HSI feature extraction and classification methods based on superpixels, there are still two significant challenges that persist.

The first challenge revolves around the initial step of extracting features from HSIs using superpixels, which involves confining the HSI to a specific band and then employing the superpixel segmentation method. It is important to note that different bands within the HSI can capture distinct attributes of the same object on the ground. This implies that by utilizing multiple bands, there exists enormous potential for more comprehensive HSI feature extraction [42]. However, effectively leveraging this potential remains a challenge that needs to be addressed.

The second challenge relates to the association between superpixel sizes and the spatial structure information they represent. Different superpixel sizes capture different levels of spatial context, providing varying amounts of spatial structure information. Existing techniques often rely on classification outcomes to determine the optimal multiscale sequence for superpixel segmentation. For example, certain methods, as demonstrated in [35], iteratively adjust the superpixel sizes based on the classification results. However, these techniques may not be practical in real-world scenarios as they lack the capability to predict the optimal multiscale ensemble in advance, resulting in a suboptimal selection of superpixel sizes.

In summary, these challenges highlight the need for further research and development in the field of HSI feature extraction using superpixels. Specifically, there is a need to explore methods that can effectively leverage the diverse attributes captured by different HSI bands and develop approaches that can predict the optimal multiscale ensemble for superpixel segmentation. Overcoming these challenges will significantly advance the accuracy and efficiency of HSI analysis and contribute to the broader field of remote sensing and image processing.

In response to the first challenge, a novel technique for extracting hyperspectral superpixel features by bands is proposed to extract rich joint spatial and spectral information from hyperspectral data. For the second challenge, an adaptive algorithm based on an explained variation (EV) [43] evaluation index is presented to determine the optimal set of multiscale superpixels. The proposed method, band-by-band adaptive multiscale superpixel feature extraction (BAMS-FE), consists of these two algorithms and has been demonstrated to perform exceptionally well not only in the top 99% principal components post-PCA transformation but also when analyzing raw, unprocessed hyperspectral data.

This study presents several innovative contributions in the field of HSI feature extraction, including the following.

1) Our study presents a band-by-band superpixel segmentation method for extracting joint spatial and spectral features from hyperspectral data, providing rich joint spatial and spectral information for each band.
2) Additionally, we introduce a new adaptive algorithm for determining the optimal multiscale superpixel set, which has the advantages of low parameter count and ease of parameter determination.
3) By combining the above two methods, the proposed BAMS algorithm demonstrates robustness to hyperparameters and initial datasets. Even with the top 99% principal components post-PCA transformation, or with raw, unprocessed hyperspectral datasets, BAMS achieves stable and satisfactory classification performance.

The rest of this article is organized as follows. In this article, we provide a comprehensive overview of the methods employed in our study, as well as a detailed description of the BAMS feature extraction algorithm proposed in Section III. The experimental design and analysis of results are presented in Section IV, followed by a summary of the key findings in Section V.

II. Related Work

Section II mainly introduces the entropy rate superpixel (ERS) segmentation method and the evaluation metrics for superpixel segmentation.

A. Entropy Rate Superpixel Segmentation

Liu et al. [44] propose a method for superpixel segmentation called the ERS segmentation method, which is based on graph theory and seeks to segment subgraphs. To evaluate the weights between graph nodes of the ERS, a random walk algorithm [45] is employed. The superpixel segmentation

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of images is achieved by using a greedy optimization algorithm [46].

ERS method can rapidly partition a grayscale image into a specified number of subimages with similar colors. Consequently, it has been widely applied in HSI processing tasks and has achieved notable results. In this article, we treat this method as a fundamental module and do not provide a detailed introduction to its specific implementation [44].

B. Explained Variation

Superpixel segmentation is a technique that involves oversegmentation, where each superpixel attempts to minimize its own internal variance, and the similarity between neighboring superpixels is not taken into account by the algorithm. As a result, the evaluation metrics for superpixel segmentation results must be designed with this characteristic in mind. Currently, based on whether actual category segmentation results are required, evaluation metrics for superpixel segmentation results can be categorized as either supervised or unsupervised. Supervised metrics include boundary recall (Rec), undersegmentation error (UE), compactness (CO), achievable segmentation accuracy (ASA), and mean distance to edge (MDE).

On the other hand, unsupervised metrics rely on the internal features of superpixels, typically the gray distribution features, and the original image pixel distribution features for evaluation. Commonly used unsupervised metrics include EV and intracluster variation (ICV). ICV quantifies the spectral variability within a superpixel, while EV measures the boundary adherence between superpixels and the raw images. In our approach, we have opted for EV as the measure for superpixel multiscale optimization methods. The formula for EV is

$$\mathrm{EV}(S) = \frac{\sum_{S_j} |S_j| (\mu(S_j) - \mu(I))^2}{\sum_{x_n} (I(x_n) - \mu(I))^2} \tag{1}$$

where $|S_j|$ represents the number of pixels within the superpixel S_j, $\mu(S_j)$ represents the spectral mean value within S_i, $\mu(I)$ represents the spectral mean value of the entire image, and $I(x_n)$ represents the gray value of x_n pixel.

III. Proposed Method

In this study, a BAMS-FE method is proposed. It addresses the challenge of capturing diverse image information across different bands and different spatial scales by adaptively selecting the optimal superpixel segmentation sequence, which leads to superior performance.

In this study, we denote an HSI as $I(H \times W \times N)$, where H, W, and N represent the height, width, and the number of spectral bands, respectively. We also denote the HSI after PCA transformation as $I_{\mathrm{pca}}(H \times W \times P)$, where P represents the number of retained principal components. Each pixel in I_{pca} is denoted as $x_i = 1, 2, \ldots, n$, where $n = H \times W$. $x_i(c)$ represents the grayscale value of the ith pixel consisting of gray values of c principal component. The initial segmentation size of the superpixel is designated as S_{size}, and the segmentation size of superpixel is recorded as $\mathrm{segsize}_{\mathrm{step}} = S_{\mathrm{size}} \times \mathrm{step}$ in each subsequent step. The superpixels in each band are denoted as $S_i = 1, 2, \ldots, m$, where

$$m = \frac{H \times W}{\mathrm{segsize}_{\mathrm{step}} \times \mathrm{segsize}_{\mathrm{step}}} \tag{2}$$

represents the number of superpixels under $\mathrm{segsize}_{\mathrm{step}}$. $|S_i|$ represents the number of pixels within the ith superpixel range.

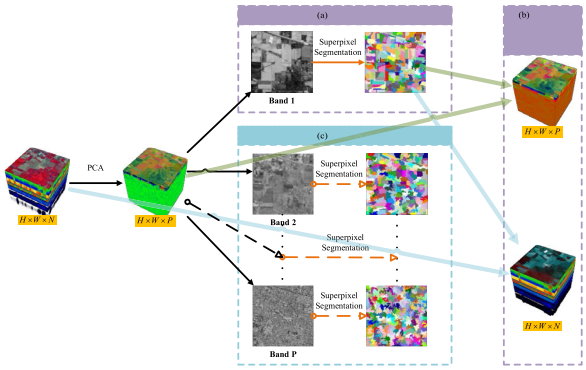


Fig. 1. Process flowchart of HSI feature extraction based on superpixels.

A. Band-by-Band Superpixel-Based Feature Extraction

The ability of superpixel segmentation algorithms to adapt to spatial features is crucial for effective HSI spatial–spectral joint feature extraction. The current mainstream approach involves applying superpixel segmentation to the first principal component after reducing dimensionality via PCA and computing the spectral mean within the superpixel range as the spectral feature of superpixel [24], [31], [35], [47], [48], [49], [50]. This approach is limited in its capacity to leverage the spatial information available in different bands of HSI, as illustrated in Fig. 1.

Therefore, the ERS segmentation method was applied to each band of an HSI in our study. This HSI may be raw data or data transformed by PCA, and the dataset to be processed can consist of one or hundreds of bands, as illustrated in Fig. 2(b). The pseudocode for the band-by-band superpixel feature extraction method is presented in Algorithm 1.

B. Adaptive Optimal Superpixel Multiscale Determination

It is important to consider that a single superpixel segmentation size may only capture features at a specific scale, which can limit the utilization of available spatial information. Therefore, combining multiple superpixel segmentation sizes in a feature extraction technique can improve results by capturing spatial information at different scales. Furthermore, current multiscale superpixel segmentation methods for HSI feature extraction and classification often lack the ability to determine an optimal set of segmentation scales beforehand, hindering their application in this field. To address this issue, this article presents an EV-based algorithm for determining the optimal multiscale set of superpixel segmentation. The proposed algorithm is illustrated in Fig. 2(b).

The pseudocode for the adaptive optimal superpixel multiscale determination method is presented in Algorithm 2.

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Algorithm 1 Band-by-Band Superpixel-Based Feature Extraction

Input: (a) HSI after performing PCA: $I_{pca}(H \times W \times P)$, (b) superpixel segmentation scale: $segsize_{step}$;
Determining the required number of superpixels for segmentation: $m = \frac{H \times W}{segsize_{step} \times segsize_{step}}$;
For $c = 1, 2, \ldots, P$
For $I_{pca}(:, :, c)$, $ERS(I_{pca}(:, :, c), m)$ superpixel segmentation is performed;
For $j = 1, 2, \ldots, m$
Compute the mean gray value within the superpixel region: $\mu(S_j)$;
Substitute the gray values of all pixels within the range S_j with $\mu(S_j)$;
End For
Obtain a gray image of superpixel segmentation features for the cth band: $I_f(c)$
End For
Output: The collection of superpixel segmentation feature grayscale images for all bands: $I_f(:)$.

C. BAMS-FE

Herein, we introduce a new approach, termed as BAMS-FE method, which combines the feature extraction method of superpixels at each band and adaptive optimal superpixel multiscale determination. The proposed method solely relies on three hyperparameters, namely, the percentage of principal components to be retained (p), the initial segmentation size of superpixels (S_{size}), and threshold value (α). The experimental results validate the robustness of these hyperparameters for feature extraction algorithms tailored for classification tasks, hence easily discernible. Algorithm 3 provides the pseudocode for the BAMS-FE method utilized in the current study. This algorithm is the BAMS algorithm for the first few principal components of PCA transformation, as shown in Fig. 2. It is noteworthy that this algorithm can accommodate various input sources, including the raw hyperspectral data sets with or without additional feature processing. In addition, we validated the robustness and versatility of this method by applying the BAMS algorithm to both the top 99% principal component after PCA transformation and the raw hyperspectral dataset without any processing in our experiment.

Algorithm 2 Adaptive Optimal Superpixel Multiscale Determination

Input: (a) HSI after performing PCA: $I_{pca}(H \times W \times P)$, (b) the initial segmentation size of the superpixel: S_{size}, (c) threshold value: α;
Initialization of the number of iterations ($step = 1$), the scale of superpixel segmentation ($segsize_{step} = S_{size} \times step$), and stopping criterion ($flag = True$);
While *flag*
Determining the required number of superpixels for segmentation: $m = \frac{H \times W}{segsize_{step} \times segsize_{step}}$;
For $c = 1, 2, \ldots, P$
For $I_{pca}(:, :, c)$, $ERS(I_{pca}(:, :, c), m)$ superpixel segmentation is performed;
For $j = 1, 2, \ldots, m$
Compute the mean gray value within the superpixel region: $\mu(S_j)$;
Substitute the gray values of all pixels within the range S_j with $\mu(S_j)$;
End For
Compute the EV value for the cth band: $EV_{band}(c) = EV(S)$;
End For
Compute the sum of the EV values for all spectral bands: $EV_{step} = sum(EV_{band}(:))$;
Record the EV_{step} in the list $EV_{steplist}$: $EV_{steplist}(step) = EV_{step}$;
If $step == 1$
$flag = True$;
Record the $segsize_{step}$: $segsize_{steplist}(step) = segsize_{step}$;
Else If $\frac{EV_{step}}{sum(EV_{steplist}(:))} < \alpha$ or $\frac{EV_{step}}{EV_{steplist}(step-1)} < 1 - \alpha$
$flag = False$
Else
Record the $segsize_{step}$: $segsize_{steplist}(step) = segsize_{step}$;
End If
End While
Output: optimal superpixel multiscale list: $segsize_{steplist}(:)$

IV. Experiments and Results

The effectiveness and robustness of the proposed method and its associated hyperparameters are evaluated through a series of experiments. Five commonly used HSI datasets are utilized to examine the impact of various hyperparameters on classification accuracy. Performance is assessed based on three metrics: overall accuracy (OA), average accuracy (AA), and the kappa coefficient.

A. Datasets and Experimental Setup

The details of the five publicly available datasets used in this study are described in the following.

1) The Indian Pines dataset was acquired using AVIRIS sensors and consists of 220 spectral bands spanning the visible to middle infrared spectrum. After excluding 20 bands reflecting water, 200 bands were utilized for the research. It has a spatial resolution of 20 m with a size of 145 × 145 pixels, encompassing 16 land cover classes.
2) The WHU-Hi-HanChuan dataset was captured using a 17-mm focal length Headwall Nano-Hyperspec imaging sensor. It comprises 274 spectral bands covering the range of 400–1000 nm. The dataset has a spatial resolution of 0.109 m and a size of 1217 × 303 pixels, containing 16 land cover classes.
3) The WHU-Hi-HongHu dataset was also captured using a 17-mm focal length Headwall Nano-Hyperspec imaging sensor. It consists of 270 spectral bands covering the

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Algorithm 3 Band-by-Band Adaptive Multiscale Superpixel Feature Extraction (BAMS-FE)

Input: (a) HSI: $I(H \times W \times N)$, (b) the percentage of the main components that need to be retained: p, (c) the initial segmentation size of the superpixel: S_{size}, (d) threshold value:α;
Perform PCA transformation on $I(H \times W \times N)$ to get $I_{pca}(H \times W \times P)$;
Initialization of the number of iterations ($step = 1$), the scale of superpixel segmentation ($segsize_{step} = S_{size} \times step$), and stopping criterion ($flag = True$);
While *flag*
  Determining the required number of superpixels for segmentation: $m = \frac{H \times W}{segsize_{step} \times segsize_{step}}$;
  For $c = 1, 2, \ldots, P$
    For $I_{pca}(:, :, c)$, $ERS(I_{pca}(:, :, c), m)$ superpixel segmentation is performed;
    For $j = 1, 2, \ldots, m$
      Compute the mean gray value within the superpixel region: $\mu(S_j)$;
      Substitute the gray values of all pixels within the range S_j with $\mu(S_j)$;
    End For
    Compute the EV value for the cth band: $EV_{band}(c) = EV(S)$;
    Obtain a gray image of superpixel segmentation features for the cth band:$I_f(c)$;
  End For
  Compute the sum of the EV values for all spectral bands: $EV_{step} = sum(EV_{band}(:))$;
  Record the EV_{step} in the list $EV_{steplist}$: $EV_{steplist}(step) = EV_{step}$;
  Obtain the collection of superpixel segmentation feature grayscale images for all bands: $I_f(:)$;
  If $step == 1$
    $flag = True$;
    Record the $segsize_{step}$: $segsize_{steplist}(step) = segsize_{step}$;
    Record the $I_f(:)$ in the list $I_{BAMS}(step)$
  Else If $\frac{EV_{step}}{sum(EV_{steplist}(:))} < \alpha$ or $\frac{EV_{step}}{EV_{steplist}(step-1)} < 1 - \alpha$
    $flag = False$
  Else
    Record the $segsize_{step}$: $segsize_{steplist}(step) = segsize_{step}$;
    Record the $I_f(:)$ in the list $I_{BAMS}(step)$;
  End If
End While
Output: Feature Extraction Dataset: $I_{BAMS}(:)$

range of 400–1000 nm. The dataset has a higher spatial resolution of 0.043 m and a size of 940 × 475 pixels. It includes 22 land cover classes.

4) The Pavia University dataset was collected using the reflective optics spectrographic image system (ROSIS). It encompasses 103 spectral bands spanning the range of 430–860 nm. The dataset has a spatial resolution of 1.3 m with a size of 610 × 340 pixels, and it consists of nine land cover classes.
5) The Salinas dataset was collected using a 224-band AVIRIS sensor. After discarding 20 invalid bands, it consists of 204 bands. The dataset has a spatial resolution of 3.7 m with a size of 512 × 217 pixels, containing 16 crop classes.

The land cover categories present in the five datasets are summarized in Tables I–V.

In this investigation, we conducted three distinct experiments to comprehensively evaluate the performance, effectiveness, and generalizability of the proposed BAMS algorithm for HSI analysis. First, we employed a grid-search approach to scrutinize the impacts of two pivotal hyperparameters of the BAMS algorithm, namely, the threshold parameter (α) and the initial scale parameter of superpixels (S_{size}), on the ultimate classification outcomes. Subsequently, we selected several state-of-the-art superpixel-based feature extraction algorithms as control experiments to benchmark the superiority of the BAMS algorithm. Moreover, to assess the versatility

TABLE I
NUMBERS OF LABELED SAMPLES OF DIFFERENT CLASSES IN INDIAN PINES DATASET

Indian Pines dataset			
Id	Class Names	Numbers	Label Colors
1	Alfalfa	46	
2	Corn-notill	1428	
3	Corn-mintill	830	
4	Corn	237	
5	Grass-pasture	483	
6	Grass-trees	730	
7	Grass-pasture-mowed	28	
8	Hay-windrowed	478	
9	Oats	20	
10	Soybean-notill	972	
11	Soybean-mintill	2455	
12	Soybean-clean	683	
13	Wheat	205	
14	Woods	1265	
15	Buildings-Grass-Trees-Drives	386	
16	Stone-Steel-Towers	93	

TABLE II
NUMBERS OF LABELED SAMPLES OF DIFFERENT CLASSES IN WHU-HI-HANCHAUN DATASET

WHU-Hi-HanChuan dataset			
Id	Class Names	Numbers	Label Colors
1	Strawberry	44735	
2	Cowpea	22753	
3	Soybean	10287	
4	Sorghum	5353	
5	Water spinach	1200	
6	Watermelon	4533	
7	Greens	5903	
8	Trees	17978	
9	Grass	9469	
10	Red roof	10516	
11	Gray roof	16911	
12	Plastic	3679	
13	Bare soil	9116	
14	Road	18560	
15	Bright object	1136	
16	Water	75401	

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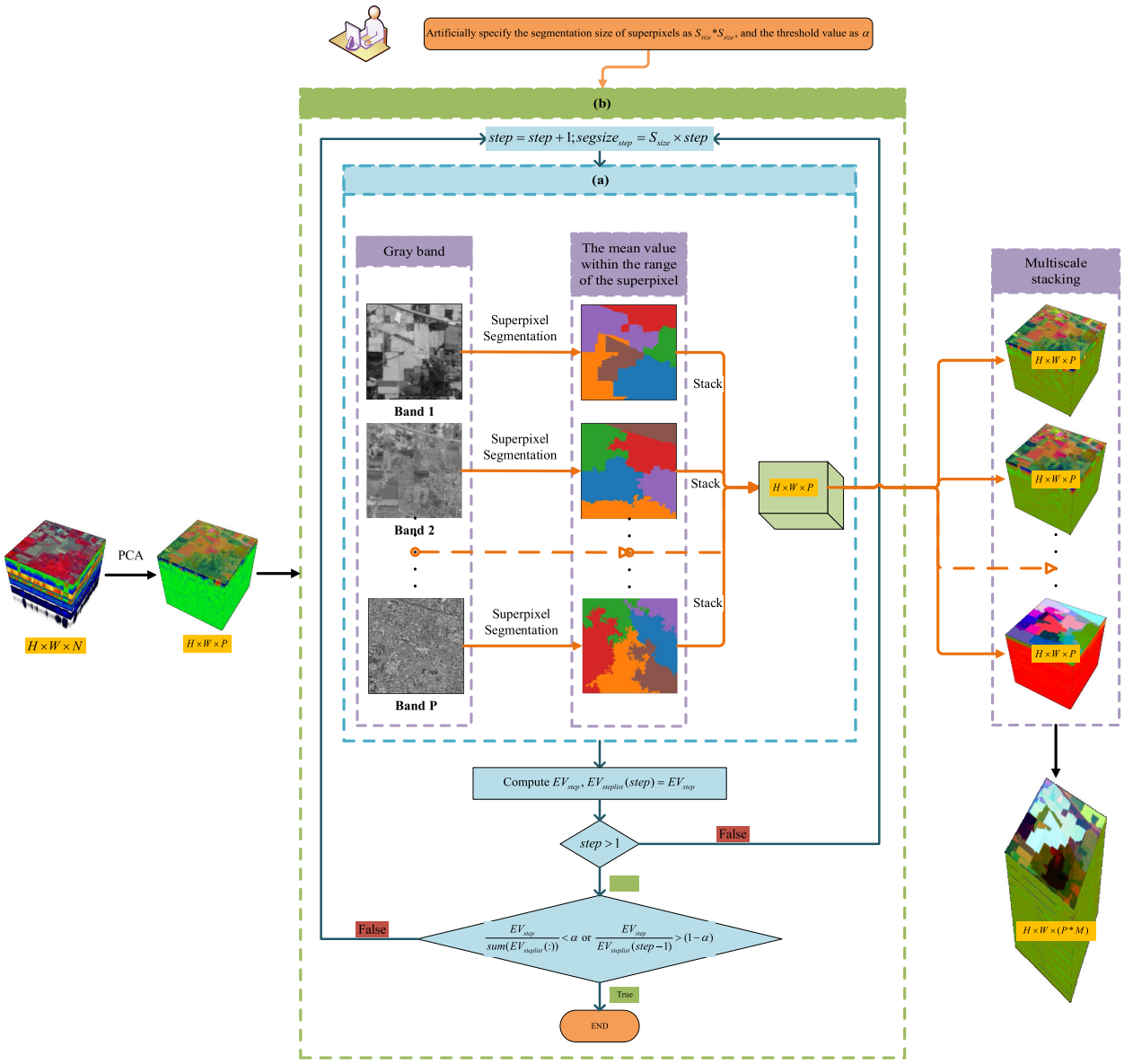


Fig. 2. BAMS algorithm flowchart.

and robustness of BAMS for HSI analysis, we performed an extensive investigation into its capability of processing images of varying dimensions. Specifically, we examined the efficacy of this algorithm on various combinations of principal components, including the first principal component, the first and second principal components, the first through third principal components, up to the first 99% of principal components, as well as on the original hyperspectral data without any PCA transformation. To evaluate the effectiveness of these feature extraction methods, we utilized an RF classifier. Furthermore, we performed a comprehensive analysis of the computational time required for various feature extraction methods.

B. Parameter Analysis

The BAMS algorithm is a multiparametric technique that relies on three key hyperparameters, p, α, and S_{size}. In view of the Indian Pines, Pavia University, Salinas, WHU-Hi-HongHu, and WHU-Hi-HanChuan datasets, where the cumulative proportion of variance explained by the first 25, four, three, six, and three principal components, respectively, reached 99%, we did not perform a grid search to compare the effects of different percentages of retained principal components on the final classification results relative to the other two parameters. Nevertheless, in Section IV-D of our article, we conducted a comparative analysis of the classification efficacy of the BAMS feature extraction algorithm under varying cumulative percentages of principal components and in the absence of PCA transformation, while keeping the parameters α and S_{size} constant.

The classification accuracy across various datasets, training sample sizes, and combinations of initial superpixel scale parameters (S_{size}) and cutoff thresholds (α) is depicted in Fig. 3. The analysis reveals intriguing patterns. For the Indian Pines, Pavia University, Salinas, WHU-Hi-HongHu,

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TABLE III

NUMBERS OF LABELED SAMPLES OF DIFFERENT CLASSES IN WHU-HI-HONGHU DATASET

WHU-Hi-HongHu			
Id	Class Names	Numbers	Label Colors
1	Red roof	14041	
2	Road	3512	
3	Bare soil	21821	
4	Cotton	163285	
5	Cotton firewood	6218	
6	Rape	44557	
7	Chinese cabbage	24103	
8	Pakchoi	4054	
9	Cabbage	10819	
10	Tuber mustard	12394	
11	Brassica parachinensis	11015	
12	Brassica chinensis	8954	
13	Small Brassica chinensis	22507	
14	Lactuca sativa	7356	
15	Celtuce	1002	
16	Film covered lettuce	7262	
17	Romaine lettuce	3010	
18	Carrot	3217	
19	White radish	8712	
20	Garlic sprout	3486	
21	Broad bean	1328	
22	Tree	4040	

TABLE IV

NUMBERS OF LABELED SAMPLES OF DIFFERENT CLASSES IN PAVIA UNIVERSITY DATASET

Pavia University dataset			
Id	Class Names	Numbers	Label Colors
1	Asphalt	6631	
2	Meadows	3682	
3	Gravel	1345	
4	Trees	947	
5	Painted metal sheets	1330	
6	Bare Soil	18649	
7	Bitumen	2099	
8	Self-Blocking Bricks	3064	
9	Shadows	3682	

TABLE V

NUMBERS OF LABELED SAMPLES OF DIFFERENT CLASSES IN SALINAS DATASET

Salinas dataset			
Id	Class Names	Numbers	Label Colors
1	Asphalt	6631	
2	Meadows	3682	
3	Gravel	1345	
4	Trees	947	
5	Painted metal sheets	1330	
6	Bare Soil	18649	
7	Bitumen	2099	
8	Self-Blocking Bricks	3064	
9	Shadows	3682	

and WHU-Hi-HanChuan datasets, setting the cutoff threshold (α) to 0.01 during training, the classification accuracy displays varying degrees of improvement. With five samples per class, the accuracy ranges from 0.11% to 3.15%, while with ten samples, it ranges from 0.03% to 2.85%, and with 15 samples, it ranges from 0.06% to 3.14%. Further examination of Fig. 3 elucidates that decreasing the cutoff threshold (α) and utilizing

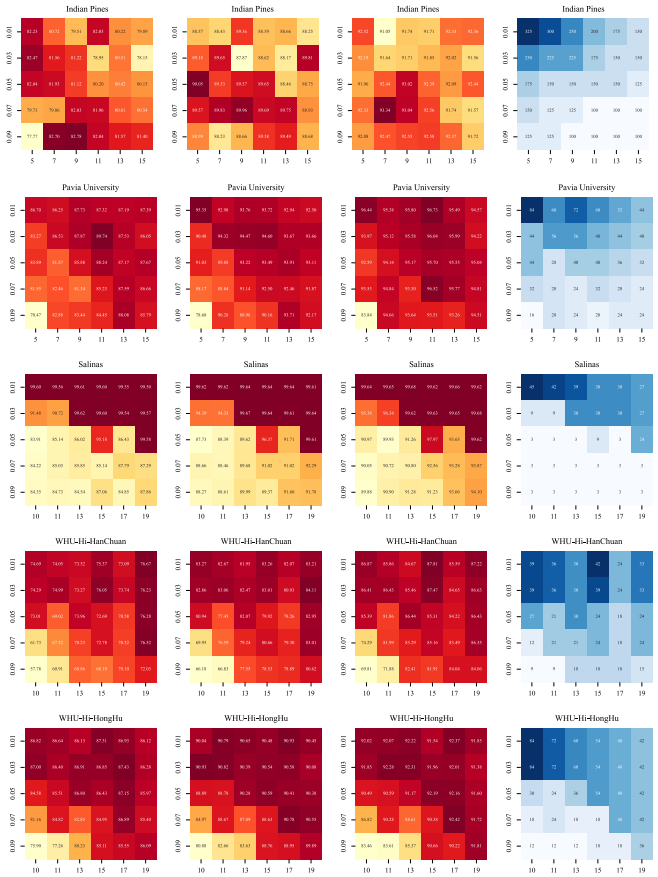


Fig. 3. The illustration provides an overview of the overall classification accuracy with respect to different datasets, S_{size} values, and α parameters. Each subplot delineates the interplay between S_{size}, positioned along the horizontal axis, and α, marked on the vertical axis. Within the figure, the first, second, and third columns depict the OA (%) achieved when five, ten, and 15 samples are randomly selected from each class for training purposes, respectively. The richness of the color palette, ranging from pale to deep red, signifies the ascending levels of classification accuracy. In the fourth column, the dimensionality of the extracted features is captured, employing a color spectrum that transitions from lighter to darker shades of navy, thereby signifying increasing dimensionality. Remarkably, the rows of the figure present data specific to the Indian Pines dataset, Pavia University dataset, Salinas dataset, WHU-Hi-HanChuan dataset, and WHU-Hi-HongHu dataset, respectively.

different initial scale parameters (S_{size}) generally enhance the classification accuracy. Moreover, as the feature datasets become more comprehensive, their dimensionality also tends to increase. These findings suggest that the extraction of multiscale features contributes to the enhancement of accuracy in HSI classification tasks. Notably, when the cutoff threshold (α) is large, augmenting the initial scale parameter (S_{size}) tends to yield an upward trend in classification accuracy. However, this trend diminishes as the cutoff threshold (α) becomes smaller.

Among the five datasets, when five samples are selected for each class during training and the cutoff threshold (α) is set to 0.01 with the minimum initial superpixel scale parameter (S_{size}), the differences in classification accuracy compared to the highest achieved accuracy using various parameter combinations are as follows: 1.48% for the Indian Pines dataset, 3.04% for the Pavia University dataset, 0.02% for the Salinas dataset, 1.98% for the WHU-Hi-HongHu dataset,

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TABLE VI

CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON INDIAN PINES DATASET WITH FIVE TRAINING SAMPLES PER CLASS (BEST ARE HIGHLIGHT IN BOLD)

Class No.	origin-RF	origin-SVM	SuperPCA-RF	SuperPCA-SVM	MSuperPCA-RF	MSuperPCA-SVM	SpaSSA-RF	SpaSSA-SVM	SHDA-RF	SHDA-SVM	SBDFE	BAMS-RF	BAMS-SVM
1	80.12±0.10	82.44±0.13	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	77.80±0.13	40.61±0.16	97.80±0.01	97.80±0.01	97.80±0.01	99.02±0.01	88.17±0.28
2	28.50±0.10	28.23±0.19	50.71±0.10	40.85±0.13	64.79±0.10	61.77±0.11	36.44±0.12	28.76±0.18	**73.72±0.10**	64.47±0.16	46.05±0.10	62.46±0.12	38.82±0.22
3	28.99±0.10	19.21±0.14	60.77±0.16	35.31±0.14	65.84±0.15	56.34±0.16	41.74±0.11	19.18±0.15	**72.78±0.17**	40.85±0.13	68.39±0.15	70.18±0.18	40.38±0.09
4	39.20±0.11	19.61±0.18	68.32±0.16	87.72±0.08	87.56±0.11	82.71±0.15	71.14±0.15	38.36±0.14	84.07±0.13	72.37±0.11	80.82±0.22	**90.95±0.09**	57.24±0.09
5	61.73±0.13	20.91±0.26	**88.77±0.09**	68.50±0.06	85.60±0.08	81.49±0.07	56.74±0.14	14.75±0.21	80.08±0.07	66.63±0.04	80.56±0.16	78.13±0.07	48.88±0.23
6	59.55±0.13	32.34±0.18	79.18±0.15	43.65±0.14	92.64±0.15	92.10±0.15	50.88±0.08	31.97±0.15	93.66±0.06	88.97±0.04	90.73±0.08	**94.52±0.09**	70.28±0.12
7	86.09±0.07	91.95±0.02	96.30±0.02	96.30±0.02	96.30±0.02	96.30±0.02	94.78±0.10	**100.00±0.00**	96.52±0.02	**100.00±0.00**	92.17±0.21	99.78±0.01	99.78±0.01
8	60.96±0.10	47.77±0.14	93.15±0.12	70.95±0.17	**100.00±0.00**	99.92±0.00	85.17±0.07	82.15±0.11	94.13±0.11	78.00±0.13	85.00±0.17	**100.00±0.00**	**100.00±0.00**
9	82.33±0.15	73.67±0.10	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	86.67±0.12	47.33±0.26	**100.00±0.00**	**100.00±0.00**	95.00±0.22	**100.00±0.00**	**100.00±0.00**
10	42.37±0.13	31.97±0.14	70.32±0.14	39.42±0.19	**83.82±0.05**	77.89±0.12	38.46±0.12	25.74±0.21	72.43±0.15	58.95±0.12	65.45±0.15	81.61±0.06	45.86±0.29
11	40.36±0.10	42.76±0.22	45.21±0.10	28.98±0.12	77.49±0.12	62.71±0.15	47.56±0.15	34.81±0.25	66.90±0.10	51.82±0.23	59.00±0.13	**79.49±0.10**	51.39±0.30
12	24.92±0.08	5.27±0.09	46.72±0.11	17.77±0.18	60.98±0.12	54.12±0.15	36.52±0.09	11.04±0.11	**80.53±0.18**	78.31±0.19	61.85±0.17	72.24±0.11	40.22±0.17
13	92.68±0.03	95.70±0.02	99.55±0.00	99.52±0.00	99.52±0.00	99.52±0.00	85.55±0.09	85.75±0.16	99.52±0.00	99.52±0.00	99.52±0.00	**99.88±0.00**	99.52±0.00
14	64.20±0.14	66.47±0.21	62.16±0.09	25.80±0.10	89.38±0.10	86.39±0.11	74.22±0.12	64.84±0.25	**94.21±0.06**	84.78±0.07	86.42±0.10	93.94±0.07	71.92±0.19
15	26.40±0.06	11.61±0.11	89.10±0.14	67.65±0.16	**93.40±0.10**	92.27±0.10	69.57±0.10	49.65±0.18	86.75±0.14	88.83±0.11	61.15±0.12	91.81±0.09	83.24±0.08
16	93.24±0.04	89.32±0.06	89.32±0.12	94.21±0.12	96.42±0.05	97.55±0.03	94.09±0.06	91.02±0.10	**100.00±0.00**	**100.00±0.00**	98.01±0.02	**100.00±0.00**	96.59±0.02
OA	44.62±3.28	37.09±3.74	62.55±2.60	41.25±4.08	79.99±2.92	73.52±4.02	52.55±4.08	37.55±5.68	79.13±2.91	67.14±5.38	68.61±4.24	**81.34±3.39**	56.59±4.89
AA	56.98±9.93	47.45±13.71	77.47±9.41	63.54±10.03	87.11±7.14	83.82±8.19	65.46±10.83	47.87±16.31	87.07±8.06	79.46±8.43	79.25±12.51	**88.38±6.34**	70.77±13.16
Kappa	38.19±3.47	29.23±3.63	58.59±2.75	35.86±4.21	77.40±3.20	70.29±4.34	47.32±4.29	30.94±5.09	76.44±3.25	63.16±5.68	64.63±4.77	**78.95±3.74**	51.06±4.90

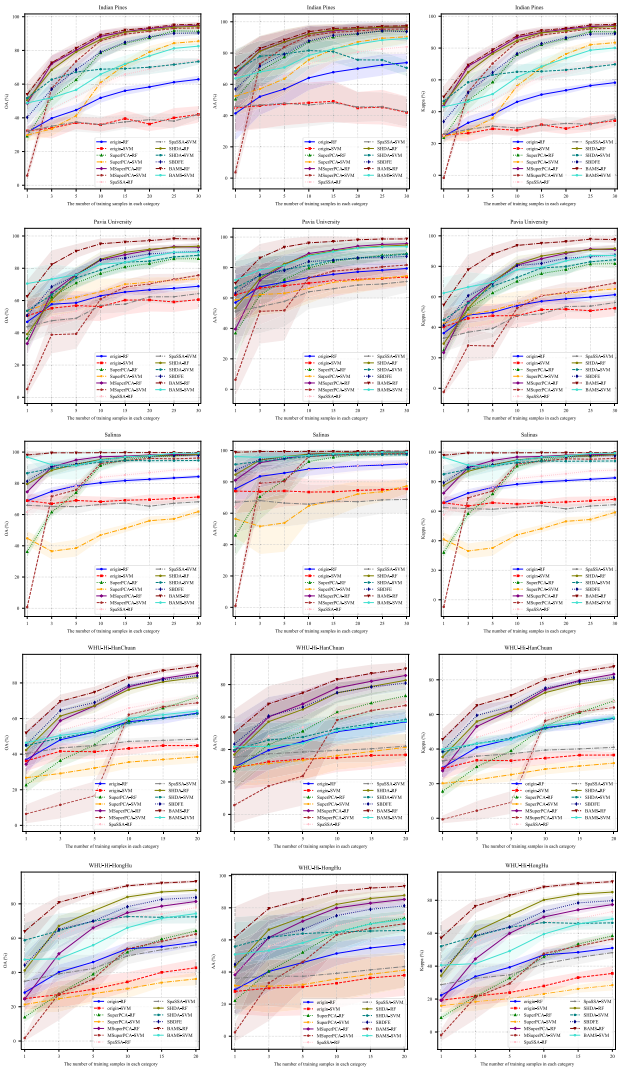


Fig. 4. Line graphs of OA, AA, and kappa for five datasets when different feature extraction methods and classifiers are employed. Each line graph represents the performance under various training set sizes, specifically selecting different numbers of samples from each class. The rows of the figure correspond to the Indian Pines, Pavia University, Salinas, WHU-Hi-HanChuan, and WHU-Hi-HongHu datasets, respectively.

and 0.61% for the WHU-Hi-HanChuan dataset. Similarly, when ten samples are selected, the differences are 0.55%, 0%, 0.05%, 0.84%, and 0.89%, while with 15 samples, the differences are 0.82%, 0.29%, 0.04%, 0.94%, and 0.3%.

Based on the analysis results and the parameter values corresponding to the highest classification accuracy observed in Fig. 3, it can be concluded that optimal classification results are often achieved with different hyperparameter combinations, varying according to the datasets and training sample sizes. Consequently, predetermining the optimal hyperparameter combination proves to be impractical. However, employing a cutoff threshold (α) of 0.01 and selecting an appropriate value for the initial superpixel scale parameter (S_{size}) offer a relatively simple and stable approach, resulting in classification accuracy only marginally lower than that of the optimal combination. Importantly, when each class utilizes ten or more samples for training, the classification accuracy under this approach is within a 1% difference compared to the optimal conditions. This demonstrates the feasibility of determining the hyperparameters α and S_{size} using the proposed method. In the upcoming experiments, for the Indian Pines, Pavia University, Salinas, WHU-Hi-HongHu, and WHU-Hi-HanChuan datasets, a cutoff threshold (α) of 0.01 and initial superpixel scale parameters (S_{size}) of five, five, ten, ten, and ten, respectively, will be employed.

C. Classification Accuracy Comparison

In this investigation, we utilized SuperPCA[1] [35] and MSuperPCA[1] [35] from 2018, along with superpixelwise adaptive SSA (SpaSSA)[2] [50], superpixel-based Brownian descriptor (SBD)[3] [49], and superpixel-level hybrid discriminant analysis (SHDA)[4] [47] algorithms from 2022 as benchmarks. These algorithms represent some of the most advanced HSI feature extraction techniques based on superpixels that have been developed in recent years.

All experiments were conducted on an Intel i7-13700k CPU. The feature extraction for SuperPCA, MSuperPCA, SpaSSA, SBD, and SHDA was implemented using MATLAB 2022b, and the classification task was performed using the RF and SVM algorithms from the sklearn library in Python 3.10. For the Indian Pines, Pavia University, and Salinas datasets, the hyperparameters of various comparative algorithms in this

[1]https://github.com/junjun-jiang/SuperPCA
[2]https://github.com/Hang-Fu/SpaSSA
[3]https://github.com/ShuzhenZhang2019/SBDFE
[4]https://github.com/ShuzhenZhang2019/SHDA

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TABLE VII

CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON PAVIA UNIVERSITY DATASET WITH FIVE TRAINING SAMPLES PER CLASS (BEST ARE HIGHLIGHT IN BOLD)

Class No.	origin-RF	origin-SVM	SuperPCA-RF	SuperPCA-SVM	MSuperPCA-RF	MSuperPCA-SVM	SpaSSA-RF	SpaSSA-SVM	SHDA-RF	SHDA-SVM	SBDFE	BAMS-RF	BAMS-SVM
1	63.83±0.05	55.45±0.16	39.50±0.08	33.45±0.14	44.39±0.08	47.76±0.14	43.86±0.14	46.52±0.15	71.90±0.09	74.20±0.11	70.28±0.10	**92.24±0.06**	83.75±0.10
2	48.17±0.13	51.75±0.29	81.46±0.08	77.80±0.18	80.24±0.08	36.48±0.23	48.53±0.11	40.83±0.22	71.48±0.12	73.41±0.18	76.66±0.13	**86.15±0.10**	71.79±0.14
3	42.09±0.13	31.26±0.23	78.55±0.08	57.54±0.12	78.25±0.11	30.37±0.26	39.37±0.10	36.18±0.23	84.60±0.12	68.10±0.07	84.28±0.11	**88.28±0.10**	40.86±0.19
4	**88.22±0.08**	87.60±0.08	32.29±0.07	19.63±0.10	60.57±0.07	31.81±0.24	65.71±0.14	63.48±0.15	68.34±0.10	49.94±0.13	58.46±0.11	85.50±0.09	82.39±0.10
5	94.85±0.08	96.49±0.05	98.65±0.01	81.70±0.08	98.25±0.02	90.05±0.11	96.21±0.04	96.76±0.03	97.12±0.03	92.92±0.03	94.40±0.04	**99.69±0.00**	98.81±0.00
6	52.78±0.11	37.77±0.30	84.26±0.07	70.76±0.18	82.40±0.10	21.00±0.19	64.35±0.14	56.52±0.22	84.26±0.11	68.20±0.19	72.89±0.22	**99.40±0.02**	61.52±0.25
7	81.10±0.10	88.78±0.10	89.80±0.06	97.98±0.02	98.40±0.04	84.10±0.27	53.63±0.11	57.66±0.17	97.97±0.04	**100.00±0.00**	86.82±0.03	**100.00±0.00**	**100.00±0.00**
8	62.97±0.14	63.55±0.26	63.55±0.09	28.48±0.29	93.39±0.06	24.83±0.33	57.81±0.12	54.93±0.28	74.07±0.12	86.45±0.07	82.00±0.15	**97.21±0.04**	95.60±0.06
9	**99.93±0.00**	99.84±0.00	80.57±0.08	99.89±0.00	99.29±0.01	99.92±0.00	70.06±0.09	66.58±0.04	91.61±0.06	94.20±0.03	76.51±0.05	92.23±0.08	97.58±0.06
OA	58.61±4.73	56.87±9.05	70.86±2.99	61.93±8.96	76.11±3.78	39.33±9.21	53.37±5.75	49.01±7.25	75.76±5.07	73.99±6.62	75.62±4.15	**90.65±4.25**	76.02±4.46
AA	70.44±9.17	68.06±16.33	72.07±6.77	63.02±12.42	81.69±6.26	51.81±19.68	59.95±11.01	57.72±16.57	82.37±8.82	78.60±9.06	78.03±10.61	**93.41±5.39**	81.37±10.01
Kappa	49.74±4.49	47.60±7.82	62.56±3.17	51.11±9.23	69.74±4.34	27.75±8.48	43.68±5.85	39.22±5.95	69.60±5.62	66.95±6.91	68.66±4.81	**88.00±5.24**	69.54±4.85

TABLE VIII

CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON SALINAS DATASET WITH ONE TRAINING SAMPLES PER CLASS (BEST ARE HIGHLIGHT IN BOLD)

Class No.	origin-RF	origin-SVM	SuperPCA-RF	SuperPCA-SVM	MSuperPCA-RF	MSuperPCA-SVM	SpaSSA-RF	SpaSSA-SVM	SHDA-RF	SHDA-SVM	SBDFE	BAMS-RF	BAMS-SVM
1	91.33±0.11	96.84±0.04	55.29±0.13	49.97±0.05	**100.00±0.00**	8.98±0.15	51.87±0.19	58.42±0.21	99.40±0.02	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
2	93.81±0.11	78.04±0.12	31.15±0.08	49.20±0.01	99.94±0.00	5.64±0.08	77.67±0.18	79.13±0.11	80.60±0.16	89.52±0.15	99.04±0.04	**100.00±0.00**	**100.00±0.00**
3	45.01±0.19	52.31±0.21	69.03±0.19	67.10±0.23	99.99±0.00	0.00±0.00	48.72±0.18	62.97±0.21	78.07±0.18	**100.00±0.00**	46.69±0.11	**100.00±0.00**	**100.00±0.00**
4	96.91±0.04	97.66±0.03	55.33±0.22	96.33±0.05	90.81±0.09	0.00±0.00	80.49±0.24	85.17±0.22	97.33±0.06	**99.93±0.00**	99.93±0.00	99.83±0.00	99.78±0.00
5	89.83±0.10	88.64±0.12	46.99±0.19	77.33±0.18	65.37±0.20	0.00±0.00	45.20±0.19	45.74±0.21	85.63±0.19	**99.33±0.00**	97.52±0.09	99.06±0.00	98.92±0.00
6	97.67±0.01	96.59±0.02	40.94±0.17	74.49±0.26	93.75±0.10	0.01±0.00	90.86±0.03	95.46±0.02	88.04±0.14	**99.92±0.00**	99.92±0.00	99.89±0.00	99.87±0.00
7	94.91±0.13	91.09±0.11	29.21±0.07	38.11±0.16	53.00±0.08	0.00±0.00	68.17±0.22	82.05±0.23	88.32±0.12	74.80±0.23	99.01±0.00	**100.00±0.00**	99.37±0.00
8	45.40±0.23	48.97±0.26	16.44±0.07	25.81±0.07	42.89±0.15	0.00±0.00	40.90±0.22	46.85±0.26	59.85±0.22	69.12±0.24	55.17±0.33	93.14±0.13	**98.85±0.00**
9	75.24±0.18	80.81±0.22	37.50±0.12	24.36±0.06	82.68±0.16	0.00±0.00	79.68±0.25	80.61±0.23	94.59±0.05	99.97±0.00	99.49±0.01	**99.98±0.00**	99.97±0.00
10	30.46±0.19	31.12±0.18	38.59±0.16	42.81±0.18	53.62±0.21	0.00±0.00	31.47±0.15	32.10±0.18	81.18±0.16	88.81±0.20	57.95±0.24	**96.61±0.02**	65.52±0.19
11	74.14±0.23	62.57±0.29	49.33±0.20	79.67±0.23	58.78±0.22	0.00±0.00	44.43±0.24	42.75±0.21	96.41±0.09	**100.00±0.00**	97.15±0.01	97.99±0.02	95.54±0.01
12	72.96±0.22	76.98±0.17	40.75±0.12	45.18±0.09	72.61±0.14	0.00±0.00	75.03±0.17	78.12±0.12	89.77±0.13	**100.00±0.00**	98.68±0.06	**100.00±0.00**	**100.00±0.00**
13	98.36±0.01	96.42±0.05	98.25±0.00	**99.41±0.01**	98.24±0.00	0.00±0.00	86.06±0.20	78.77±0.20	92.24±0.14	97.92±0.00	97.82±0.00	98.03±0.00	98.03±0.00
14	85.43±0.05	84.83±0.07	54.59±0.13	54.37±0.12	96.31±0.03	0.00±0.00	85.88±0.18	81.68±0.21	80.80±0.26	90.84±0.20	94.20±0.08	**97.95±0.01**	96.35±0.00
15	56.53±0.23	59.64±0.22	31.90±0.12	34.83±0.16	89.86±0.14	0.00±0.00	67.05±0.28	74.90±0.30	72.40±0.19	79.95±0.20	72.82±0.30	**99.52±0.00**	99.10±0.00
16	61.22±0.23	43.68±0.15	39.36±0.12	43.67±0.08	99.07±0.00	0.00±0.00	52.64±0.24	47.68±0.17	68.65±0.21	68.01±0.14	80.45±0.29	**100.00±0.00**	86.57±0.19
OA	68.81±3.34	68.95±4.52	36.21±3.32	44.79±4.53	74.51±5.55	0.72±0.72	61.53±4.18	65.93±4.97	79.59±4.30	86.48±5.62	81.29±6.64	**98.13±2.66**	96.80±1.10
AA	75.58±14.19	74.14±14.19	45.92±13.06	56.41±12.13	81.06±9.40	0.91±1.49	64.13±19.88	67.02±19.34	84.58±14.50	91.13±8.59	87.24±9.69	**98.87±1.14**	96.12±2.48
Kappa	65.65±3.51	65.72±4.71	32.04±3.25	40.94±4.74	72.25±5.87	-4.80±0.93	57.66±4.56	62.43±5.41	77.41±4.72	85.01±6.17	79.27±7.31	**97.92±2.94**	96.43±1.23

TABLE IX

CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON WHU-HI-HANGCHUAN DATASET WITH TEN TRAINING SAMPLES PER CLASS (BEST ARE HIGHLIGHT IN BOLD)

Class No.	origin-RF	origin-SVM	SuperPCA-RF	SuperPCA-SVM	MSuperPCA-RF	MSuperPCA-SVM	SpaSSA-RF	SpaSSA-SVM	SHDA-RF	SHDA-SVM	SBDFE	BAMS-RF	BAMS-SVM
1	55.72±0.06	48.80±0.13	42.67±0.07	43.74±0.09	65.93±0.08	54.29±0.07	69.49±0.08	70.06±0.02	79.68±0.05	62.34±0.06	81.43±0.08	**84.46±0.07**	63.78±0.05
2	27.85±0.06	19.26±0.10	33.75±0.05	20.26±0.12	53.40±0.05	33.21±0.16	33.11±0.08	4.61±0.05	45.84±0.06	30.80±0.05	**53.71±0.07**	50.46±0.08	30.18±0.07
3	43.23±0.10	10.36±0.13	77.51±0.11	43.79±0.14	83.75±0.07	58.15±0.10	66.07±0.07	32.90±0.20	87.80±0.05	76.68±0.07	85.87±0.05	**88.84±0.06**	38.99±0.09
4	77.84±0.09	73.42±0.11	96.35±0.02	89.05±0.04	98.21±0.01	91.16±0.04	85.09±0.07	84.71±0.10	95.54±0.01	95.13±0.01	98.04±0.00	97.70±0.02	**98.22±0.00**
5	62.66±0.11	82.18±0.04	99.70±0.00	97.17±0.03	99.51±0.00	95.87±0.02	88.00±0.07	97.67±0.03	97.09±0.03	93.43±0.04	92.73±0.03	92.90±0.05	**99.88±0.00**
6	17.94±0.06	2.95±0.05	64.16±0.11	4.06±0.06	80.96±0.05	48.15±0.09	33.89±0.06	4.49±0.05	60.02±0.10	29.93±0.11	50.87±0.09	**91.20±0.06**	20.88±0.10
7	78.12±0.06	68.68±0.12	64.76±0.09	55.75±0.10	88.62±0.06	74.12±0.10	82.71±0.07	82.98±0.11	86.80±0.07	73.55±0.10	78.63±0.06	**95.55±0.04**	63.22±0.22
8	36.77±0.06	7.80±0.08	37.37±0.08	3.02±0.03	66.80±0.08	13.87±0.09	45.88±0.06	10.11±0.08	52.98±0.08	41.93±0.09	61.26±0.06	**80.58±0.05**	40.55±0.08
9	16.74±0.06	0.93±0.03	47.49±0.08	3.41±0.05	67.10±0.07	16.65±0.09	34.42±0.06	0.27±0.01	56.72±0.09	24.92±0.08	51.79±0.08	**74.13±0.06**	21.18±0.14
10	52.24±0.09	1.91±0.04	56.69±0.09	4.16±0.08	69.65±0.09	25.14±0.11	61.21±0.08	2.70±0.05	83.36±0.06	30.76±0.06	**86.92±0.04**	83.40±0.07	8.66±0.11
11	73.60±0.12	84.08±0.16	76.08±0.11	48.94±0.12	94.79±0.05	96.86±0.05	70.28±0.10	84.80±0.17	93.32±0.07	**99.60±0.01**	86.54±0.09	86.75±0.07	39.23±0.17
12	37.06±0.08	6.06±0.07	78.94±0.02	41.80±0.25	91.28±0.08	**96.46±0.05**	60.71±0.06	5.41±0.04	77.99±0.10	15.08±0.15	84.99±0.09	94.24±0.07	60.58±0.18
13	35.24±0.09	4.27±0.07	57.12±0.08	9.01±0.11	68.71±0.09	51.67±0.19	48.32±0.07	6.15±0.05	63.65±0.06	23.99±0.24	61.94±0.06	**74.64±0.08**	53.76±0.10
14	46.95±0.09	13.25±0.12	38.16±0.08	17.67±0.03	50.73±0.09	23.65±0.08	45.26±0.09	11.32±0.10	52.43±0.09	15.65±0.11	52.48±0.08	**63.77±0.09**	29.85±0.09
15	69.75±0.04	59.53±0.07	60.29±0.05	48.09±0.05	70.76±0.08	53.10±0.10	72.99±0.08	59.64±0.09	74.10±0.08	54.86±0.10	76.98±0.05	**77.45±0.06**	57.87±0.07
16	84.43±0.10	73.18±0.07	76.36±0.08	49.98±0.10	97.13±0.03	97.39±0.01	88.20±0.06	72.74±0.07	90.74±0.07	75.45±0.02	96.27±0.05	94.06±0.06	**98.17±0.03**
OA	58.22±2.43	43.19±3.59	58.76±3.18	35.54±3.48	77.82±1.83	62.20±2.22	65.88±2.20	47.16±1.66	76.24±2.95	57.41±2.06	78.62±2.23	**82.86±2.26**	60.09±1.61
AA	51.01±7.95	34.79±8.63	62.96±7.05	36.24±8.80	77.96±6.10	58.11±8.48	61.60±7.08	39.41±7.51	74.88±6.65	52.76±8.01	75.03±6.26	**83.13±6.19**	51.56±9.36
Kappa	52.34±2.42	34.74±3.80	53.37±3.30	28.42±3.11	74.34±2.02	56.36±2.47	60.86±2.30	39.40±2.07	72.61±3.22	51.82±2.21	75.22±2.47	**80.15±2.51**	53.39±1.72

experiment are set based on the existing configurations found in relevant research articles and open-source code. In the case of the WHU-Hi-HanChuan and WHU-Hi-HongHu datasets, the manually specified hyperparameters are set to match those used for the Salinas dataset.

We integrated Python 3.10 and MATLAB 2022b to implement the BAMS algorithm and conducted classification experiments utilizing the RF and SVM algorithms from the sklearn library. The default parameters were used for the RF and SVM classifiers in Python. After analyzing the parameters, we fixed p at 99%, α at 0.01, and S_{size} at 5×5 for Indian Pines and Pavia University datasets and 10×10 for Salinas, WHU-Hi-HongHu, and WHU-Hi-HanChuan datasets. For the Indian Pines, Pavia University, and Salinas datasets, we employ a random selection of one, three, five, ten, 15, 20, 25, and 30 samples per class for training purposes. Similarly, for the WHU-Hi-HanChuan and WHU-Hi-HongHu datasets, we adopt a random selection of one, three, five, ten, 15, and 20 samples per class for training. We varied the random seed over time.

When the number of samples in a class was insufficient for the specified number of training samples, we randomly selected 50% of the samples from that class for training. We performed 20 repetitions of each method for feature extraction and random selection of training sets and calculated the OA, AA, kappa coefficient, and corresponding standard deviations to compare the performance of each method.

The proposed algorithm was thoroughly evaluated against various state-of-the-art hyperspectral feature extraction algorithms for classification. The comparative analysis is visually

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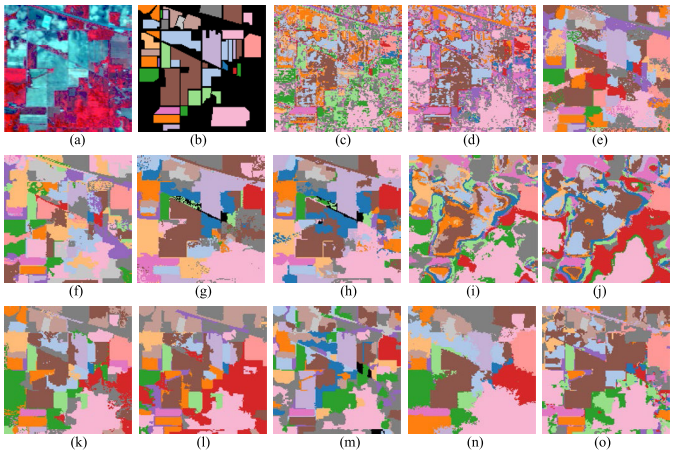


Fig. 5. Classification results of different feature extraction methods and classifiers in the Indian Pines dataset when five samples from each class were selected for training. (a) Pseudo-color image and (b) ground truth. The classification results are represented by the OA values as follows: (c) 0.478, (d) 0.413, (e) 0.655, (f) 0.416, (g) 0.825, (h) 0.761, (i) 0.564, (j) 0.392, (k) 0.832, (l) 0.688, (m) 0.643, (n) 0.861, and (o) 0.587.

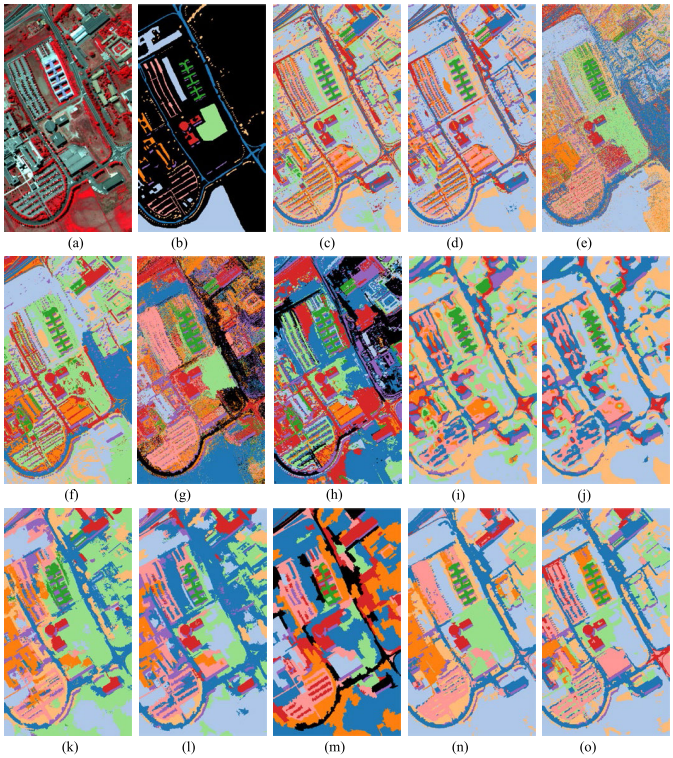


Fig. 6. Classification results of different feature extraction methods and classifiers in the Pavia University dataset when five samples from each class were selected for training. (a) Pseudo-color image and (b) ground truth. The classification results are represented by the OA values as follows: (c) 0.627, (d) 0.665, (e) 0.732, (f) 0.53, (g) 0.78, (h) 0.479, (i) 0.636, (j) 0.596, (k) 0.734, (l) 0.756, (m) 0.7, (n) 0.95, and (o) 0.796.

depicted in the line chart presented in Fig. 4. Additionally, the results are visually demonstrated through a series of images in Figs. 5–9, accompanied by detailed performance metrics in Tables VI–X.

Based on the experimental outcomes, it was observed that the RF classifier employed in the BAMS algorithm consistently outperformed the SVM classifier.

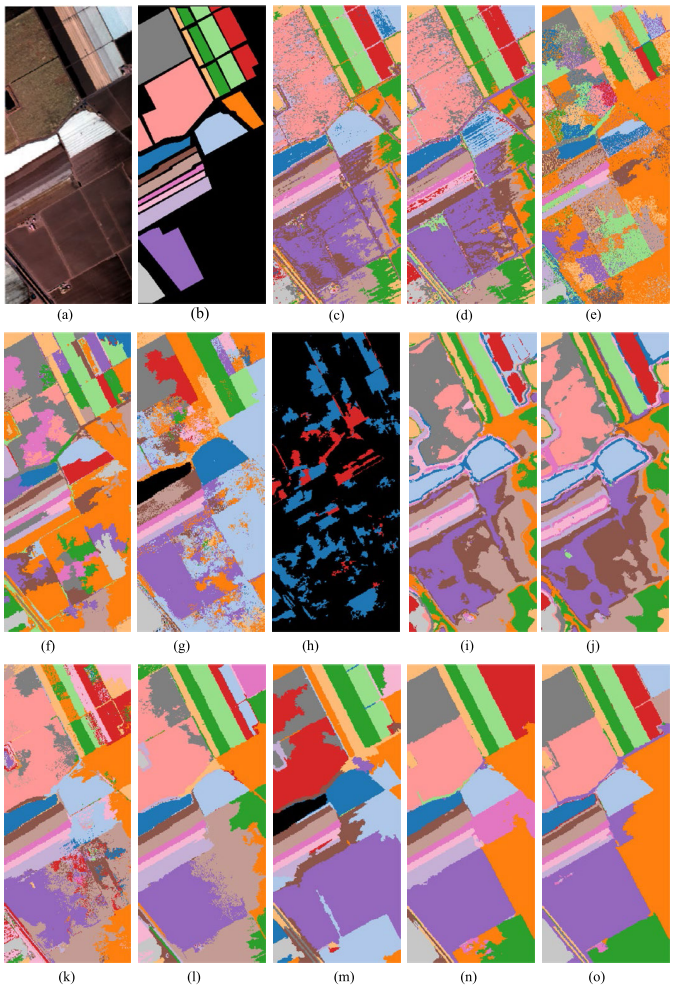


Fig. 7. Classification results of different feature extraction methods and classifiers in the Salinas dataset when one sample from each class was selected for training. (a) Pseudo-color image and (b) ground truth. The classification results are represented by the OA values as follows: (c) 0.7, (d) 0.715, (e) 0.359, (f) 0.456, (g) 0.619, (h) 0.017, (i) 0.642, (j) 0.656, (k) 0.716, (l) 0.872, (m) 0.858, (n) 0.993, and (o) 0.951. The classification failure in case (h) can be attributed to the insufficient number of training samples, which renders the SVM classifier incapable of executing decision-level voting effectively. Consequently, the classifier proves to be ineffective in this scenario.

Specifically, when examining the Indian Pines dataset under different scenarios where the training sets were constructed by selecting one, three, five, ten, 15, 20, 25, or 30 samples per class, the proposed feature extraction method exhibited superior performance in conjunction with the RF classifier. Notably, it surpassed other state-of-the-art techniques by achieving the highest OA values in both RF and SVM classification. The margins of improvement amounted to 2.17%, 0.49%, 1.35%, 0.83%, 1.06%, 0.42%, 0.85%, and 0.78% when compared to the alternative methods, respectively.

In the case of the Pavia University dataset, the margins of improvement over other techniques were significantly higher, amounting to 8.94%, 13.74%, 14.89%, 9.59%, 6.45%, 5.41%, 4.96%, and 4.46%. Similarly, for the Salinas dataset, the proposed algorithm achieved notable margins of improvement of 11.65%, 8.48%, 4.62%, 2.54%, 2.29%, 1.49%, 1.31%, and 0.94%. The WHU-Hi-HanChuan dataset exhibited consistent performance enhancements, with margins of 6.78%, 5.15%,

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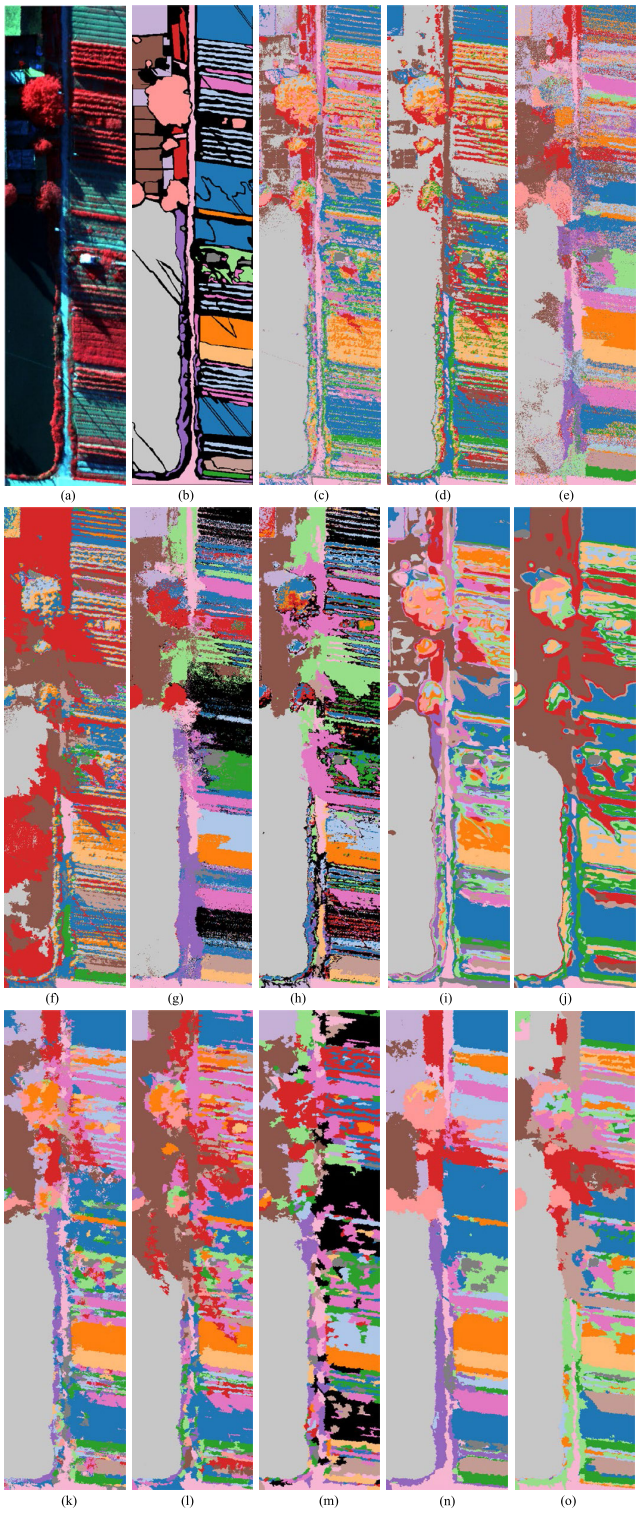


Fig. 8. Classification results of different feature extraction methods and classifiers in the WHU-Hi-HanChuan dataset when ten samples from each class were selected for training. (a) Pseudo-color image and (b) ground truth. The classification results are represented by the OA values as follows: (c) 0.607, (d) 0.508, (e) 0.643, (f) 0.315, (g) 0.807, (h) 0.65, (i) 0.693, (j) 0.507, (k) 0.777, (l) 0.605, (m) 0.774, (n) 0.853, and (o) 0.603.

5.82%, 4.24%, 4.47%, and 3.81% achieved when selecting one, three, five, ten, 15, or 20 samples per class for the training set. Finally, for the WHU-Hi-HongHu dataset, the proposed algorithm attained margins of improvement of 5.16%, 15.76%, 16.39%, 6.51%, 5.23%, and 5.16% when utilizing one, three, five, ten, 15, or 20 samples per class for training purposes.

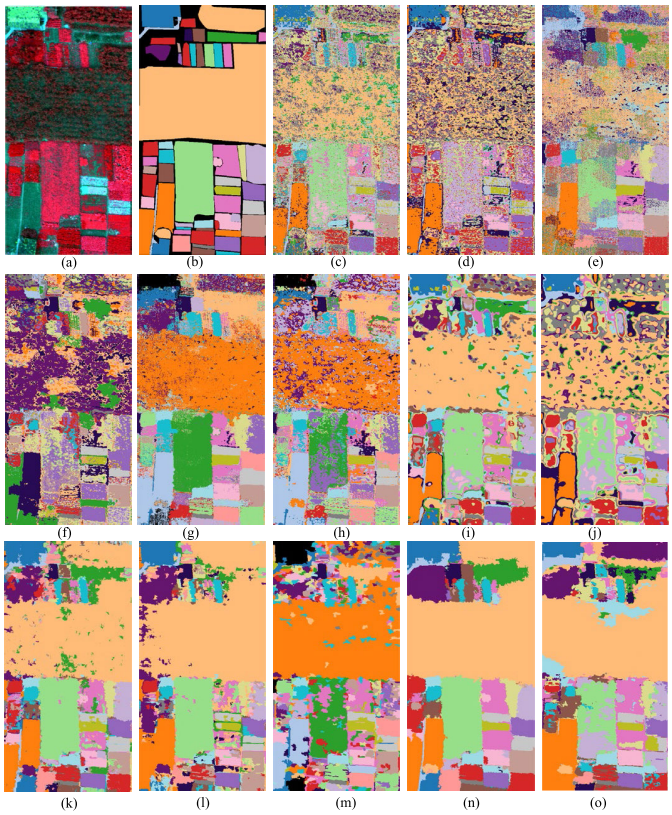


Fig. 9. Classification results of different feature extraction methods and classifiers in the WHU-Hi-HongHu dataset when ten samples from each class were selected for training. (a) Pseudo-color image and (b) ground truth. The classification results are represented by the OA values as follows: (c) 0.534, (d) 0.367, (e) 0.518, (f) 0.223, (g) 0.744, (h) 0.557, (i) 0.727, (j) 0.546, (k) 0.86, (l) 0.765, (m) 0.758, (n) 0.91, and (o) 0.689.

D. Discussion

Section IV-B presented the impact of varying parameters α and S_{size} on feature extraction efficacy using the BAMS algorithm when applied to principal components with a cumulative eigenvalue of 99%. Herein, we fixed α to 0.01 and S_{size} to 5 or 10 and performed feature extraction on different cumulative principal components. Moreover, we applied the BAMS feature extraction on unprocessed hyperspectral data and utilized an RF classifier to classify the extracted features, comparing the outcomes with previous experiments.

The common findings revealed in Fig. 10 exhibit the following characteristics: (1) With a decrease in the size of the training set, higher feature dimensions contribute to enhanced classification accuracy. However, this trend becomes less pronounced as the training set size increases. (2) Utilizing the first principal component alone as a classification feature fails to provide effective discriminative information. At least two principal components are required to achieve highly accurate classification tasks. (3) A feature dimension of up to ten often yields satisfactory classification results, and increasing the feature dimension does not necessarily guarantee improved classification performance.

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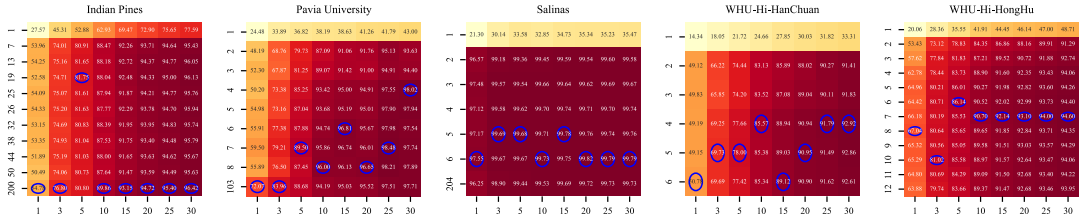


Fig. 10. Impact of preserving varying numbers of principal components on the resultant classification outcomes. x-axis represents the number of training samples selected per class, while the y-axis signifies the retained number of principal components post-PCA transformation, and the blue circles indicate the positions where the maximum values occur in each column. Notably, the final row in the figure portrays the outcomes when employing the BAMS algorithm directly on the original data for Indian Pines, Pavia University, and Salinas dataset.

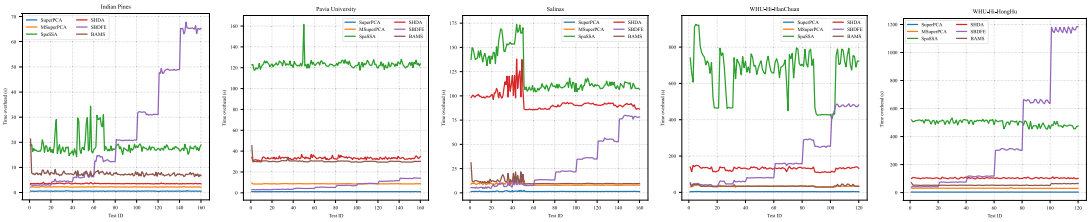


Fig. 11. Graph of time expenditure for different feature extraction algorithms in each run. The experiments presented in Fig. 4 serve as the basis for this graphical representation. Notably, the training set size is dynamically adjusted every 20 iterations to align with the corresponding values illustrated in Fig. 4.

For the Indian Pines dataset, the classification accuracy differences obtained when retaining 99% of the principal component information, compared to the best classification accuracy (excluding the data in the last row of Fig. 10), are as follows: 0.24%, 0.13%, 0.11%, 0.53%, 0.85%, 0.16%, 0.23%, and 0.37%. In the case of the Pavia University dataset (excluding the data in the last row of Fig. 10), the respective values are: 9.3%, 5.83%, 4.25%, 2.58%, 1.81%, 1.74%, 0.93%, and 0%. Similarly, for the Salinas dataset (excluding the data in the last row of Fig. 10), the values are: 0.07%, 0.12%, 0.14%, 0.07%, 0.14%, 0.2%, 0.1%, and 0.12%. Regarding the WHU-Hi-ChanChuan dataset, the respective differences are: 0.95%, 3.88%, 3.8%, 2.05%, 2.04%, 1.91%, 1.86%, and 1.09%. Finally, for the WHU-Hi-HongHu dataset, the values are: 2.42%, 0.31%, 0%, 0.18%, 0.12%, 0.11%, 0.27%, and 0.2%.

These statistics demonstrate that retaining 99% of the principal component information already achieves highly accurate classification (with the difference to the best classification accuracy mostly within 1%) for the Indian Pines, Salinas, and WHU-Hi-HongHu datasets. However, for the Pavia University and WHU-Hi-HanChuan datasets, retaining a larger proportion of the principal component information contributes to improved classification accuracy.

Considering the spatial resolutions of the five datasets (Indian Pines: 20 m, Salinas: 3.7 m, WHU-Hi-HongHu: 0.043 m, Pavia University: 1.3 m, and WHU-Hi-HanChuan: 0.109 m) and their imaging characteristics (Indian Pines and Salinas exhibit high spatial resolutions, contain vegetation regions with substantial spectral variability, and experience more noise, while WHU-Hi-HongHu demonstrates high spatial resolution resulting in significant spectral variability between pixels; Pavia University predominantly encompasses urban areas with buildings and artificial objects, showing relatively smaller spectral variability compared to agricultural regions), we conjecture that for pixels with higher noise levels, retaining 99% of the principal component information already achieves favorable classification results. Conversely, for pixels with lower noise levels, greater feature information assists in further enhancing classification accuracy. However, the validity of this conclusion necessitates additional experiments for validation.

This discussion not only underscores the applicability of the BAMS approach for feature extraction across various dimensions, ranging from one to several hundred, but also highlights the potential for further improvements in the method, such as determining the optimal feature dimension and addressing related aspects.

In addition, we conducted a comparison of the computational time associated with several different feature extraction methods. It is important to note that the most time-consuming aspect among these methods is the process of superpixel segmentation. However, it is worth mentioning that all of these algorithms employ the ERS algorithm and utilize the C++ code provided by Mingyu Liu, which can be integrated into MATLAB. Furthermore, it should be emphasized that all the comparative algorithms were executed in MATLAB 2022b without any parallel computing procedures.

Regarding the BAMS algorithm, we employed parallel computing with 12 threads in MATLAB for the calculation of the mean within the superpixel region and the computation of EV values. On the other hand, the SBD algorithm operates at a superpixel level, incorporating the time cost of the nearest neighbor classification algorithm when estimating the computational time for statistical feature extraction. The ultimate results depicting the computational time are presented in Fig. 11.

It is worth mentioning that the recorded time expense curves in Fig. 11 were obtained while striving to minimize additional computational burdens. Nevertheless, some unavoidable disturbances can introduce fluctuations in the time expense curves, often leading to abrupt changes.

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TABLE X
CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON WHU-HI-HONGHU DATASET WITH TEN TRAINING SAMPLES PER CLASS (BEST ARE HIGHLIGHT IN BOLD)

Class No.	origin-RF	origin-SVM	SuperPCA-RF	SuperPCA-SVM	MSuperPCA-RF	MSuperPCA-SVM	SpaSSA-RF	SpaSSA-SVM	SHDA-RF	SHDA-SVM	SBDFE	BAMS-RF	BAMS-SVM
1	64.40±0.05	40.49±0.14	42.34±0.08	9.03±0.08	72.52±0.06	60.04±0.07	76.86±0.07	50.35±0.11	87.00±0.10	74.43±0.06	**89.74±0.06**	88.78±0.11	63.36±0.09
2	62.90±0.07	51.98±0.07	46.76±0.10	12.48±0.14	65.38±0.10	42.29±0.13	68.42±0.09	36.96±0.06	74.81±0.10	33.23±0.17	69.52±0.07	**87.21±0.10**	48.74±0.13
3	71.13±0.06	77.96±0.19	78.05±0.07	12.57±0.18	93.20±0.02	89.00±0.05	75.76±0.04	83.70±0.08	86.92±0.02	88.43±0.00	84.78±0.07	**93.68±0.05**	89.68±0.00
4	56.72±0.08	35.22±0.12	46.46±0.07	37.99±0.12	74.77±0.04	48.68±0.10	78.98±0.08	63.64±0.09	90.37±0.03	85.52±0.14	88.40±0.06	**96.31±0.02**	75.16±0.05
5	48.45±0.08	21.49±0.18	64.87±0.10	51.68±0.17	93.72±0.03	87.44±0.06	70.76±0.10	25.87±0.24	91.48±0.03	89.26±0.16	90.12±0.03	**98.55±0.02**	43.85±0.36
6	63.99±0.08	37.13±0.21	48.50±0.07	23.73±0.19	83.24±0.06	63.79±0.11	80.20±0.05	38.53±0.22	90.55±0.02	90.98±0.01	78.97±0.06	**92.74±0.02**	66.36±0.11
7	38.19±0.08	25.51±0.18	31.63±0.07	18.66±0.08	48.96±0.10	31.14±0.11	49.43±0.09	32.70±0.20	65.18±0.08	26.64±0.13	53.04±0.12	**68.48±0.07**	31.92±0.16
8	14.85±0.05	6.52±0.06	60.71±0.06	12.94±0.09	77.63±0.06	54.64±0.04	30.88±0.06	9.71±0.09	73.01±0.08	47.60±0.09	49.76±0.10	**96.65±0.04**	68.80±0.13
9	71.94±0.08	57.40±0.21	80.09±0.10	60.20±0.24	89.80±0.03	69.79±0.14	82.99±0.09	69.01±0.22	87.85±0.04	83.24±0.03	92.11±0.02	**93.90±0.02**	78.17±0.21
10	20.24±0.07	13.65±0.16	56.28±0.10	24.36±0.22	65.18±0.08	44.31±0.18	54.65±0.09	33.61±0.24	**78.90±0.05**	64.23±0.08	38.58±0.11	74.56±0.05	30.64±0.21
11	17.43±0.07	10.15±0.13	51.16±0.08	18.48±0.11	61.82±0.09	34.74±0.09	46.69±0.07	13.39±0.13	67.76±0.08	43.31±0.08	52.29±0.06	**78.49±0.07**	44.16±0.19
12	47.36±0.09	17.16±0.20	71.36±0.04	45.50±0.31	77.80±0.05	68.87±0.07	58.59±0.05	16.71±0.22	63.54±0.06	25.99±0.23	62.04±0.09	**84.15±0.09**	29.65±0.30
13	41.24±0.08	19.09±0.13	37.51±0.05	5.92±0.05	54.66±0.04	34.90±0.06	53.00±0.08	23.83±0.12	58.26±0.07	27.34±0.05	51.05±0.10	**75.30±0.06**	57.92±0.10
14	47.57±0.06	33.07±0.21	51.15±0.10	19.37±0.23	63.97±0.05	10.47±0.19	69.56±0.07	37.29±0.19	75.63±0.09	29.10±0.25	74.17±0.05	**85.48±0.08**	53.99±0.11
15	67.46±0.06	52.85±0.15	91.34±0.00	76.96±0.11	93.95±0.01	91.56±0.01	78.24±0.07	74.12±0.12	89.21±0.05	86.61±0.02	84.18±0.05	**99.18±0.00**	95.03±0.01
16	61.18±0.09	53.65±0.18	82.98±0.08	60.74±0.12	**97.96±0.00**	89.84±0.09	75.75±0.08	69.54±0.13	92.29±0.03	82.81±0.11	90.32±0.06	97.14±0.02	69.07±0.15
17	62.08±0.10	42.58±0.21	97.02±0.01	57.79±0.07	**97.94±0.01**	92.52±0.03	72.65±0.10	40.84±0.20	88.88±0.08	59.26±0.26	87.17±0.06	94.18±0.03	77.31±0.22
18	46.01±0.08	22.23±0.15	81.52±0.07	46.83±0.15	90.38±0.06	69.99±0.10	69.02±0.07	49.39±0.16	90.51±0.06	78.09±0.08	71.99±0.13	**96.03±0.05**	72.01±0.17
19	48.89±0.06	10.90±0.11	56.96±0.07	23.47±0.14	66.24±0.09	49.20±0.16	72.86±0.07	7.03±0.06	70.85±0.06	57.87±0.07	72.15±0.04	**84.64±0.06**	49.37±0.14
20	63.42±0.09	7.20±0.07	80.39±0.07	6.86±0.13	94.03±0.08	79.34±0.15	84.95±0.07	4.46±0.08	93.44±0.07	59.65±0.20	93.37±0.05	**98.50±0.02**	78.73±0.22
21	58.77±0.08	54.18±0.19	98.81±0.01	**100.00±0.00**	99.90±0.00	99.41±0.01	78.27±0.08	42.93±0.32	98.14±0.02	99.85±0.00	99.85±0.00	**100.00±0.00**	99.76±0.00
22	59.74±0.09	35.41±0.18	68.41±0.09	60.30±0.11	94.33±0.02	75.91±0.08	86.13±0.08	37.80±0.26	92.65±0.06	93.66±0.04	78.16±0.05	**99.66±0.01**	99.19±0.00
OA	53.81±3.93	34.50±5.66	51.82±3.54	30.56±5.44	74.85±1.80	53.78±4.13	72.33±3.44	49.60±4.83	84.01±1.47	72.61±6.06	78.32±2.88	**90.52±0.88**	66.06±2.66
AA	51.54±7.53	32.99±15.70	64.74±6.76	35.72±13.83	79.88±4.92	63.08±9.24	68.85±7.59	39.15±16.11	82.15±5.92	64.87±10.18	75.08±6.49	**90.16±4.47**	64.67±14.05
Kappa	46.85±3.56	27.96±4.56	45.64±3.26	23.15±4.26	69.85±1.94	47.59±3.54	66.73±3.61	41.44±4.65	80.20±1.67	66.50±5.96	73.31±3.21	**88.11±1.08**	59.52±2.84

Fig. 11 illustrates a distinct pattern in the time expense of various feature extraction algorithms. In the experiment, the time expense of the BAMS algorithm rapidly decreases after the first execution and remains at a low level in subsequent iterations. This can be attributed to the initial overhead required for MATLAB to establish a multithreaded pool during the first execution, resulting in higher time expense. In terms of time efficiency, the BAMS algorithm consistently outperforms the SpaSSA algorithm in subsequent comparisons. Additionally, the time expense of the SBD algorithm exhibits a direct correlation with the increase in training set size, making the BAMS algorithm the superior choice in most scenarios. Notably, as the image size increases, the SHDA algorithm experiences a rapid escalation in time expense. Consequently, the BAMS algorithm outperforms the SHDA algorithm in terms of time expense for datasets other than Indian Pines. However, it is important to note that the time expense of the BAMS algorithm falls short when compared to the SuperPCA and MSuperPCA algorithms.

V. CONCLUSION

This article presents a novel approach for feature extraction from HSIs through a band-by-band adaptive multiscale superpixel algorithm. The proposed method leverages the spatial structural information of different spectral bands by implementing a per-band superpixel segmentation technique. To address the challenge of selecting an optimal set of superpixels across multiple scales, an unsupervised approach based on the EV evaluation metric is proposed. These two algorithms together constitute the proposed BAMS feature extraction algorithm. Comparative analysis with the state-of-the-art superpixel-based feature extraction algorithms demonstrates that BAMS has several advantages, including fewer parameters, higher adaptability, and improved classification accuracy. The proposed approach can be employed as a preprocessing method for HSIs ranging from several bands to hundreds of bands and has a wide range of applications in further image analysis.

The methodology proposed in this study holds promise for further advancements in the determination of optimal parameters. Additionally, exploring the integration of this method's underlying principles into the field of deep learning presents an intriguing direction for further investigation.

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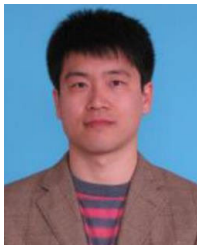
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